

NITESH NARESH NARWADE

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Education

IIITD, Delhi	Dec 2020 – Dec 2022
Master of Technology CB	CGPA: 7.85
Expertise Area	
- Computer Science - Multithreading - Data Structures and Algorithm - Mathematics	- Software Development - HFT - STL (C++) - Calculus
- Memory management - Debugging (GDB) - Advanced C++ - Trading	- Low Latency - Linux - OOPS Concepts C++ - Git

Experience

KIFS Trading Capital (High Frequency Trading Firm), Mumbai, India	Feb 2024 – Presently Working
Software Developer (C++)	(9 months)

- Works on the market data coming from the exchange (NSE). Where write C++ code from the scratch and use the relevant data structures and STL to stores the Live Market data.
- My primary work is here to developed the infrastructure (in C++) and see the feeder part (data receiving) and if any changes has come from the NSE needs to implement those changes in infrastructure code.
- Works on the Broadcast data (Transcode 7208, and 7202), where data was coming from NSE in coded buffer. Where I have to decode the buffer by LZ0 algorithm. And after using the swapendians I needed to decode the buffer and make available the data.
- I need to use multithreading for request and response order. I need to works on the Shared memory (SHM) where n number of clients can connect and access required orderbook or market data.
- While testing code If core has generated then I have to debug it by GDB tool. Starting the project from the scratch, integrating libraries (C++) with the project, building own static libraries (C++) for the project, building the project by the make file, and testing the build binary on the production server. All these are coming under my work.
- Working on the Linux environment, familiar with all the required Linux command, keep eye on the CPU performance by using htop command. Making projects in Linux with vim editor and maintaining the project structure (build file, make file, source file, and include file).
- Works on the Tick-by-Tick data (FNO), needs to write Low Latency code so that no tick should be miss. Works with the cache memory and utilise it as more as it possible (for low latency code).
- Proper knowledge of OOPS concepts, STL, and Data Structures (in C++). Also needs to deals with pointers, smart pointers, using heap section of the memory for memory management.
- Works on the TCP, UDP protocol. Kernel tunning and low-level system programming.
- Works with the Git. Add, remove, commit, pull, and push from the Git.

Dolat Capital (High frequency Trading Firm), Mumbai, India	Dec 2022 – Feb 2024
Software Developer (C++)	(14 months)

- Working on Options. Like BANKNIFTY, FINNIFTY, and etc. Using data structures and Algorithm (in C++) while writing Execution code.
- Writing C++ (Low Latency) Code in OPPS concepts while building trading Strategies. Low Latency Memory Management and can write code while cache memory keeps in mind.
- Code and understand the algorithm of AES_256_GCM for security purpose. Writing Execution from scratch in C++ and test the respective code.
- Can write multithreading code and optimize the memory.
- Solving Quants issues on live trading and on simulation by debugging Code. Working on Linux OS and very well-known with Linux commands.
- Managing a team of four people (Junior developers).

IIITD-Cosy lab	Aug 2021 – Dec 2022
(Researcher Java Programmer)	(16 months)

- Build a real time food object detection Android Application (WhatDish). Application detect food and shows the nutritional values of the food.
- Trained 61 thousand of food images on YOLOv4 and YOLOv5.

Projects

Solace Recovery

- Works on the solace recovery (Snap short recovery), Where I have phrased the hexadecimal data in human readable format and code everything in C++ to works properly.
- Previously we used to do TCP recovery which used to cost lots of time to recover market data. But with this solace recovery (snapshot recovery) now we have optimised the recovery time.
- Technology used in this project is UDP. File handling in C++, used proper Data Structures, STL, reading parameters from the config file, and write code in OOPS with proper memory management in C++.

Make Available Broadcast data (Transcodes: 7208 and 7202)

- In the HFT (High Frequency Trading firm) NSE continuous send the UDP packets of market data (Tick by Tick data). Collecting all those Tick-by-Tick market data without missing single Tick is a challenge. Because one tick has come in micro seconds.
- So, in this project I have collect the data in proper data structure (in queue) and make required operation on it (making order book).
- Here I have used shared memory concepts to send the data on shared memory and any client can access that data from SHM.
- File handling, used proper data structures, STL, reading parameters from the config file, and write code in OOPS with proper memory management in C++.

Calculating VWAP (Volume Weighted Average Price)

- In this project VWAP has calculated for current market data. And dumped the calculated VWAP and Market Data in one file.
- In this project I have used function pointer, proper send the market data request and collect the Market data response.
- Made the proper connection between the NSE server such that I can read provide stream from NSE server.
- File handling, reading parameters form the config file, used proper data structures, used shared memory concept, and memory management in C++.

Technical Skills

Languages: C++, Python, Java, Bash.

Developer Tools: Linux, GDB, Vim editor, GCC compiler.

Technologies: Multithreading, Memory Management, Low Latency, Smart Pointer, Pointers and References in C++.